Dynamic rotation with $\dot{\mathbf{j}}$ and $\mathbf{h}$ retained: equations and iterative forward solution for an elastic Earth with an ocean

Working notes (prepared with Claude) — to be checked

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Abstract

The main note (`rotation_dynamic_elastic.pdf`, "N") drops the rate of change of the inertia perturbation, $\dot{\mathbf{j}}$, from the Liouville equation. This note solves a narrower problem in which $\dot{\mathbf{j}}$ is kept, together with a prescribed relative angular momentum $\mathbf{h}(t)$ and external torque $\boldsymbol{\Gamma}(t)$. The Earth is elastic, the ocean is hydrostatic, and there is no fluid core and no damping. The inertia perturbation is written $\mathbf{j} = \mathsf{A}\sigma + B\boldsymbol{\omega}$, so that $\dot{\mathbf{j}} = \mathsf{A}\dot{\sigma} + B\dot{\boldsymbol{\omega}}$. The key practical point is that the whole left-hand side of the Liouville equation is the time derivative of the angular momentum $\boldsymbol{\ell} = I_1\boldsymbol{\omega} + \mathbf{j} + \mathbf{h}$. Stepping $\boldsymbol{\ell}$ in time therefore keeps $\dot{\mathbf{j}}$ exactly while never requiring $\dot{\mathbf{j}}$, or $\dot{\sigma}$, to be computed. The resulting solver is the iterative scheme of N§8.2 with the rotation update replaced, and $\mathbf{q}$ no longer used.

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1 Scope and assumptions

The problem is the same as in N: given a direct load $\zeta(\mathbf{x}, t)$, find the sea-level change ξ, displacement $\mathbf{u}$, gravitational potential perturbation ϕ and rotation perturbation $\boldsymbol{\omega}$. The assumptions are:

(A1) **Elastic, quasi-static deformation**, with spherically symmetric Love numbers, as in N(A1).

(A2) **Hydrostatic ocean on fixed shorelines**, as in N(A2). The total load is $\sigma = \zeta + \rho_w C\xi$, with mass conserved.

(A3) **Prescribed relative angular momentum $\mathbf{h}(t)$ and external torque $\boldsymbol{\Gamma}(t)$.** Either may be zero. Displacements are measured in a Tisserand frame of the solid Earth.

(A4) **No fluid core and no damping.** These are left for later.

(A5) $I_1 = I_2$, and only the equatorial problem is solved. The axial relation is noted in §2.

The one change of physics relative to N is that $\dot{\mathbf{j}}$ (and $\dot{\mathbf{h}}$) are kept. Notation follows N: $\mathbf{I}$ is the background inertia tensor, $\mathbf{c}$ its perturbation, $\mathbf{j} = \mathbf{c}\boldsymbol{\Omega}$, $\boldsymbol{\Omega} = \Omega\hat{\mathbf{e}}_3$, and $\mathsf{E} = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, with $\mathsf{E}^2 = -1$ and $\mathsf{E}^{-1} = -\mathsf{E}$. Where no confusion arises, $\boldsymbol{\omega}$, $\mathbf{j}$, $\mathbf{h}$, $\boldsymbol{\Gamma}$ denote their equatorial 2-vectors.

2 The Liouville equations

In the Earth-fixed frame, the angular momentum $\mathbf{L} = (\mathbf{I}+\mathbf{c})(\boldsymbol{\Omega}+\boldsymbol{\omega})+\mathbf{h}$ satisfies $\dot{\mathbf{L}}+(\boldsymbol{\Omega}+\boldsymbol{\omega})\times\mathbf{L} = \boldsymbol{\Gamma}$. Linearising, with $\mathbf{I}\boldsymbol{\Omega} = I_3\boldsymbol{\Omega}$ (N eq. 8):

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(\mathbf{I}\boldsymbol{\omega} + \mathbf{j} + \mathbf{h}\big) + \boldsymbol{\Omega} \times \big(\mathbf{I}\boldsymbol{\omega} + \mathbf{j} + \mathbf{h}\big) + \boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\Omega}) = \boldsymbol{\Gamma}. \tag{1}$$

For $I_1 = I_2$ the equatorial components take the 2-vector form (N eq. 10)

$$\boxed{I_1\dot{\boldsymbol{\omega}} + \dot{\mathbf{j}} + \dot{\mathbf{h}} = \Omega\mathsf{E}\big[(I_3 - I_1)\boldsymbol{\omega} - \mathbf{j} - \mathbf{h}\big] + \boldsymbol{\Gamma}} \tag{2}$$

or, equivalently, the excitation form (N eq. 11)

$$\dot{\boldsymbol{\omega}} = \sigma_{\mathrm{e}}\mathsf{E}(\boldsymbol{\omega} - \boldsymbol{\chi}), \qquad \boldsymbol{\chi} = \frac{1}{I_3 - I_1}\Big[\mathbf{j} + \mathbf{h} + \frac{1}{\Omega}\mathsf{E}\big(\boldsymbol{\Gamma} - \dot{\mathbf{j}} - \dot{\mathbf{h}}\big)\Big], \qquad \sigma_{\mathrm{e}} = \frac{\Omega(I_3 - I_1)}{I_1}. \tag{3}$$

Nothing is dropped in eq. (2): it is the exact linearisation.

The angular-momentum form. Define the equatorial angular-momentum perturbation and the "torque" term

$$\boldsymbol{\ell} := I_1\boldsymbol{\omega} + \mathbf{j} + \mathbf{h}, \qquad \mathbf{T} := \Omega\mathsf{E}\big[(I_3 - I_1)\boldsymbol{\omega} - \mathbf{j} - \mathbf{h}\big]. \tag{4}$$

The left side of eq. (2) is exactly $\dot{\boldsymbol{\ell}}$, so

$$\boxed{\dot{\boldsymbol{\ell}} = \mathbf{T} + \boldsymbol{\Gamma}, \qquad \boldsymbol{\omega} = \frac{\boldsymbol{\ell} - \mathbf{j} - \mathbf{h}}{I_1}} \tag{5}$$

Since $\boldsymbol{\ell} = I_1\boldsymbol{\omega} + \mathbf{j} + \mathbf{h}$, $\mathbf{T}$ can also be written $\Omega\mathsf{E}(I_3\boldsymbol{\omega} - \boldsymbol{\ell})$. That form is algebraically equal but numerically poor: $I_3\boldsymbol{\omega}$ and $\boldsymbol{\ell}$ are each about $1/\delta \approx 300$ times larger than their difference. Always evaluate $\mathbf{T}$ as in eq. (4).

Static limit. Setting $\dot{\boldsymbol{\omega}} = \dot{\mathbf{j}} = \dot{\mathbf{h}} = \mathbf{0}$ and $\boldsymbol{\Gamma} = \mathbf{0}$ gives $(I_3 - I_1)\boldsymbol{\omega} = \mathbf{j} + \mathbf{h}$: AA24's static relation with $\mathbf{h} = -\mathbf{k}$.

Axial component. The third component of eq. (1) is $I_3\dot{\omega}_3 + \dot{j}_3 + \dot{h}_3 = \Gamma_3$, i.e. $\ell_3 := I_3\omega_3 + j_3 + h_3$ satisfies $\dot{\ell}_3 = \Gamma_3$. It has no coupling to the equatorial problem and, from rest with no torque, gives the algebraic $I_3\omega_3 = -(j_3 + h_3)$. It is not considered further.

3 The inertia perturbation: $\mathbf{j} = \mathsf{A}\sigma + B\boldsymbol{\omega}$

From the potential. By MacCullagh's formula (N§6.1), the equatorial inertia perturbation is proportional to the degree-2, order-1 coefficients of the gravitational potential perturbation at radius b:

$$\mathbf{j} = \Omega(c_{13}, c_{23}) = \frac{\Omega b^3}{\sqrt{12\pi/5}\,G}\,\boldsymbol{\phi}_{21} = (I_3 - I_1)\beta\,\boldsymbol{\phi}_{21}, \qquad \beta = \sqrt{\tfrac{5}{12\pi}}\frac{\Omega b^3}{G(I_3 - I_1)}. \tag{6}$$

Here $\boldsymbol{\phi}_{21}$ is the potential of all perturbed mass, but not ψ itself.

From the loads. For an elastic Earth, $\boldsymbol{\phi}_{21}$ has two sources: the total surface load (through AA24's load Love number k_2, which includes the load's own potential) and the centrifugal perturbation $\boldsymbol{\psi}_{21} = r\boldsymbol{\omega}$ (through the tidal Love number k_2^T):

$$\boldsymbol{\phi}_{21} = k_2\,\boldsymbol{\sigma}_{21} + k_2^T r\,\boldsymbol{\omega}, \qquad r = \sqrt{\tfrac{4\pi}{15}}\,\Omega b^2. \tag{7}$$

Substituting into eq. (6),

$$\boxed{\mathbf{j} = \mathsf{A}\sigma + B\boldsymbol{\omega}, \qquad \mathsf{A}\sigma := (I_3 - I_1)\beta k_2\,\boldsymbol{\sigma}_{21}, \qquad B := (I_3 - I_1)\beta r k_2^T = (I_3 - I_1)\kappa} \tag{8}$$

with $\kappa = k_2^T/k_s = 0.314$ (N eq. 19). A is linear in σ and picks out its degree-2, order-1 part; B is a scalar. Both are constant in time because the Earth is elastic, so

$$\dot{\mathbf{j}} = \mathsf{A}\dot{\sigma} + B\dot{\boldsymbol{\omega}}. \tag{9}$$

Here σ is the *total* load, ocean included.

The ocean. $\sigma = \zeta + \rho_w C\xi$, and ξ depends on σ and on $\boldsymbol{\omega}$ through the sea-level equation. So eq. (8) is exact, but σ itself is an unknown. Eliminating it (by linearity) gives the reduced form

$$\mathbf{j} = \mathbf{j}^{\zeta} + (I_3 - I_1)\mathsf{K}\boldsymbol{\omega}, \tag{10}$$

where $\mathbf{j}^{\zeta}$ is the inertia perturbation from the direct load with the ocean responding and $\boldsymbol{\omega} = \mathbf{0}$, and K is the feedback matrix of N§6.2 (solid Earth plus ocean pole tide). Without an ocean, $\mathsf{K} = \kappa\mathsf{1}$ and $\mathbf{j}^{\zeta} = \mathsf{A}\zeta$. Eq. (10) is used for the analysis and tests below; the solver itself uses eq. (8).

Interpretation. In eq. (2), $I_1\dot{\boldsymbol{\omega}} + \dot{\mathbf{j}} = (I_1 + B)\dot{\boldsymbol{\omega}} + \mathsf{A}\dot{\sigma}$. The deformation raised by the wobble adds $B = (I_3 - I_1)\kappa$ to the wobble's effective inertia, and $\mathsf{A}\dot{\sigma}$ is the angular momentum carried by mass redistribution. Relative to I_1, B is $\delta\kappa \approx 1.0 \times 10^{-3}$.

4 Closed equations in continuous time

4.1 The $\dot{\boldsymbol{\omega}}$ form, and iterating on $\dot{\mathbf{j}}$

Substituting eq. (9) into eq. (2):

$$I_1\dot{\boldsymbol{\omega}} = \Omega\mathsf{E}\big[(I_3 - I_1)\boldsymbol{\omega} - \mathbf{j} - \mathbf{h}\big] + \boldsymbol{\Gamma} - \dot{\mathbf{h}} - \mathsf{A}\dot{\sigma} - B\dot{\boldsymbol{\omega}}. \tag{11}$$

$\dot{\boldsymbol{\omega}}$ appears on both sides, through $B\dot{\boldsymbol{\omega}}$. It can be found by iteration: start with $\dot{\mathbf{j}}^{(0)} = \mathbf{0}$, compute $\dot{\boldsymbol{\omega}}^{(k)}$ from eq. (2), set $\dot{\mathbf{j}}^{(k+1)} = \mathsf{A}\dot{\sigma} + B\dot{\boldsymbol{\omega}}^{(k)}$, and repeat. Each iteration multiplies the error by $B/I_1 = \delta\kappa \approx 1.0 \times 10^{-3}$, so it converges in two or three iterations. The limit is the closed form

$$(I_1 + B)\,\dot{\boldsymbol{\omega}} = \Omega\mathsf{E}\big[(I_3 - I_1)\boldsymbol{\omega} - \mathbf{j} - \mathbf{h}\big] + \boldsymbol{\Gamma} - \dot{\mathbf{h}} - \mathsf{A}\dot{\sigma}. \tag{12}$$

The difficulty with the ocean. This route needs $\dot\sigma = \dot\zeta + \rho_w C \dot\xi$. $\dot\zeta$ is prescribed, but $\dot\xi$ is the rate of change of the sea-level solution itself, which would have to be obtained by differentiating the solution in time. The next subsection avoids it.

4.2 The $\boldsymbol{\ell}$ form

Eq. (5) contains the same physics: differentiating $\boldsymbol{\ell} = I_1\boldsymbol{\omega} + \mathbf{j} + \mathbf{h}$ and using eq. (9) reproduces eq. (11). But $\dot{\mathbf{j}}$ never appears explicitly. It is carried inside $\dot{\boldsymbol{\ell}}$, and only $\mathbf{j}$ itself, at a given instant, is needed to recover $\boldsymbol{\omega}$ from $\boldsymbol{\ell}$. With eq. (8), that recovery is

$$\boldsymbol{\omega} = \frac{\boldsymbol{\ell} - \mathsf{A}\sigma - \mathbf{h}}{I_1 + B} \tag{13}$$

when σ is known. With the ocean, σ depends on $\boldsymbol{\omega}$, which is resolved by the same iteration as the static solver (§6).

4.3 Free wobble with $\dot{\mathbf{j}}$ kept

With eq. (10), $\mathbf{j}^\zeta = \mathbf{h} = \boldsymbol{\Gamma} = \mathbf{0}$, eq. (2) becomes

$$\big[I_1 1 + (I_3 - I_1)\mathsf{K}\big]\dot{\boldsymbol{\omega}} = \Omega(I_3 - I_1)\,\mathsf{E}(1 - \mathsf{K})\boldsymbol{\omega}. \tag{14}$$

The matrix in square brackets is the effective inertia. It and $1 - \mathsf{K}$ share the eigenvectors of K, so the same argument as in N§6.3 gives a purely oscillatory free motion with

$$\sigma'_{\mathrm{CW}} = \Omega(I_3 - I_1)\sqrt{\frac{\det(1-\mathsf{K})}{\det[I_1 1 + (I_3 - I_1)\mathsf{K}]}} \xrightarrow{\mathsf{K}=\kappa 1} \frac{\Omega(I_3 - I_1 - B)}{I_1 + B}. \tag{15}$$

Keeping $\dot{\mathbf{j}}$ lengthens the Chandler period by about half a day (§8).

5 Time discretisation

5.1 The step

Integrate eq. (5) from t_n to $t_{n+1} = t_n + \Delta t$. The left side integrates *exactly*:

$$\boldsymbol{\ell}_{n+1} - \boldsymbol{\ell}_n = \int_{t_n}^{t_{n+1}} (\mathbf{T} + \boldsymbol{\Gamma})\,\mathrm{d}t. \tag{16}$$

Only the right side is approximated, by the trapezoidal rule. With $s := \Delta t/2$:

$$\boldsymbol{\ell}_{n+1} = \boldsymbol{\ell}_n + s\big(\mathbf{T}_n + \mathbf{T}_{n+1}\big) + s\big(\boldsymbol{\Gamma}_n + \boldsymbol{\Gamma}_{n+1}\big). \tag{17}$$

This is the point of the $\boldsymbol{\ell}$ form: the $\dot{\mathbf{j}}$ term of eq. (2) becomes exactly $\mathbf{j}_{n+1} - \mathbf{j}_n$. So $\dot{\mathbf{j}}$ is kept in full, but neither $\dot{\mathbf{j}}$ nor $\dot\sigma$ is ever computed.

Solving for $\boldsymbol{\omega}_{n+1}$. Substitute $\boldsymbol{\ell}_{n+1} = I_1\boldsymbol{\omega}_{n+1} + \mathbf{j}_{n+1} + \mathbf{h}_{n+1}$ and $\mathbf{T}_{n+1} = \Omega\mathsf{E}[(I_3 - I_1)\boldsymbol{\omega}_{n+1} - \mathbf{j}_{n+1} - \mathbf{h}_{n+1}]$ into eq. (17), and collect the $\boldsymbol{\omega}_{n+1}$ terms on the left:

$$\boxed{\mathsf{P}\,\boldsymbol{\omega}_{n+1} = \mathbf{b}_n - \mathsf{C}\,\mathbf{j}_{n+1}, \qquad \mathsf{P} := I_1 1 - s\Omega(I_3 - I_1)\mathsf{E}, \quad \mathsf{C} := 1 + s\Omega\mathsf{E},} \tag{18}$$

with the known right-hand side

$$\mathbf{b}_n := \boldsymbol{\ell}_n + s\mathbf{T}_n + s\big(\boldsymbol{\Gamma}_n + \boldsymbol{\Gamma}_{n+1}\big) - \mathsf{C}\,\mathbf{h}_{n+1}. \tag{19}$$

P is invertible, since $\mathsf{P}^{-1} = [I_1 1 + s\Omega(I_3 - I_1)\mathsf{E}]/[I_1^2 + s^2\Omega^2(I_3 - I_1)^2]$. Only $\mathbf{h}$ at the time levels is needed, not $\dot{\mathbf{h}}$.

Coupling through $\mathbf{j}_{n+1}$. $\mathbf{j}_{n+1}$ depends on $\boldsymbol{\omega}_{n+1}$, both directly ($B\boldsymbol{\omega}_{n+1}$) and through the ocean's response to ψ. Eq. (18) is therefore solved iteratively, inside the sea-level loop (§6). With eq. (10), the rotational part of that iteration is $\boldsymbol{\omega} \mapsto \mathsf{P}^{-1}[\mathbf{b}_n - \mathsf{C}\mathbf{j}^{\zeta}_{n+1} - (I_3 - I_1)\mathsf{C}\mathsf{K}\boldsymbol{\omega}]$. Its contraction factor, computed for the present-day K, is

Δt	1 d	7 d	30.4 d	100 d	365 d	$\to \infty$
no ocean ($\mathsf{K} = \kappa 1$)	0.0034	0.023	0.094	0.23	0.30	0.31
present-day ocean	0.0039	0.026	0.11	0.26	0.35	0.36

For short steps it is about $\delta\kappa\, s\Omega$; for long steps it tends to the static solver's value $\approx \kappa$. In a test with weekly steps, the iteration reached a relative change of 10^{-15} in about 10 iterations, and agreed with a direct 2×2 solve to 10^{-14}.

5.2 Why the trapezoidal rule

Stepping with the current derivative only (explicit Euler, "$\boldsymbol{\omega}_{n+1} = \boldsymbol{\omega}_n + \Delta t\, \dot{\boldsymbol{\omega}}_n$") would amplify the free wobble at every step, by $\sqrt{1 + (\sigma_{\mathrm{CW}}\Delta t)^2}$. That is a factor of 1.36 per Chandler period with weekly steps and 3.5 with monthly steps. The trapezoidal rule preserves the wobble's amplitude (N§8.1), with a period error of order $(\sigma_{\mathrm{CW}}\Delta t)^2/12$. In discrete form it also conserves angular momentum exactly: eq. (17) holds to rounding (checked to 2×10^{-16}).

6 The iterative algorithm

This is N§8.2 (Scheme 1) with the rotation update replaced. $\mathbf{q}$ is not used. At each step, given $\boldsymbol{\ell}_n$, $\mathbf{T}_n$, $\boldsymbol{\omega}_n$, the direct load ζ_{n+1}, and the prescribed $\mathbf{h}_{n+1}$, $\boldsymbol{\Gamma}_n$, $\boldsymbol{\Gamma}_{n+1}$:

1. Form $\mathbf{b}_n = \boldsymbol{\ell}_n + s\mathbf{T}_n + s(\boldsymbol{\Gamma}_n + \boldsymbol{\Gamma}_{n+1}) - \mathsf{C}\mathbf{h}_{n+1}$. Initialise $\boldsymbol{\omega}^{(0)} = \boldsymbol{\omega}_n$ and the load $\sigma^{(0)}$ (warm start from σ_n).
2. For $k = 0, 1, \ldots$ until the load converges:
   (a) $\boldsymbol{\psi}_{21} = r\boldsymbol{\omega}^{(k)}$;
   (b) $u_{lm} = h_l\sigma^{(k)}_{lm}$ and $\phi_{lm} = k_l\sigma^{(k)}_{lm}$, adding $h^T_2\boldsymbol{\psi}_{21}$ and $k^T_2\boldsymbol{\psi}_{21}$ at $(2,1)$;
   (c) $\mathbf{j}^{(k)} = (I_3 - I_1)\beta\,\boldsymbol{\phi}_{21}$, eq. (6), with ϕ excluding ψ;
   (d) **rotation update:** $\boldsymbol{\omega}^{(k+1)} = \mathsf{P}^{-1}\big(\mathbf{b}_n - \mathsf{C}\mathbf{j}^{(k)}\big)$, eq. (18);
   (e) add $r\boldsymbol{\omega}^{(k+1)}$ to $\boldsymbol{\phi}_{21}$, form ξ (N eq. 4), and set $\sigma^{(k+1)} = \zeta_{n+1} + \rho_w C\xi$, with mass conserved.
3. Store $\boldsymbol{\omega}_{n+1} = \boldsymbol{\omega}^{(k+1)}$, $\mathbf{j}_{n+1} = \mathbf{j}^{(k)}$, $\boldsymbol{\ell}_{n+1} = I_1\boldsymbol{\omega}_{n+1} + \mathbf{j}_{n+1} + \mathbf{h}_{n+1}$ and $\mathbf{T}_{n+1} = \Omega\mathsf{E}[(I_3 - I_1)\boldsymbol{\omega}_{n+1} - \mathbf{j}_{n+1} - \mathbf{h}_{n+1}]$.

In `pyslfp`. This is a copy of `solve_generalised_equation` with the rotation update replaced. With `i` $= \beta$ (`inertia_factor`), `dI` $= I_3 - I_1$ (`polar_moment_of_inertia` − `equatorial_moment_of_inertia`), and `P`, `C`, `b_n` built from the formulas above:

```
j = dI * i * potential_change_lm.coeffs[:, 2, 1]
angular_velocity_change = np.linalg.solve(P, b_n - C @ j)
```

The `angular_momentum_change` argument is not used: $\mathbf{h}$ enters through $\mathbf{b}_n$. `pyslfp` is non-dimensional, so Δt, $\mathbf{h}$ and $\boldsymbol{\Gamma}$ must be expressed in its units (`time_scale` etc.). As noted in N§6.2, the sidereal Ω should be used.

Convergence. The rotational part contracts as in the table of §5. The ocean-load part is unchanged from the static solver, which dominates the iteration count.

7 Initial conditions

- **From rest:** $\zeta(0) = 0$, $\mathbf{h}_0 = \mathbf{0}$, $\boldsymbol{\omega}_0 = \mathbf{0}$, so $\mathbf{j}_0 = \mathbf{0}$, $\boldsymbol{\ell}_0 = \mathbf{0}$, $\mathbf{T}_0 = \mathbf{0}$.
- **From equilibrium** with a pre-existing load ζ_0 and $\mathbf{h}_0$: run the static solver with `angular_momentum_change` $= -\mathbf{h}_0$ (AA24's $\mathbf{k} = -\mathbf{h}$). This gives $\boldsymbol{\omega}_0$ and $\mathbf{j}_0$; then $\boldsymbol{\ell}_0 = I_1\boldsymbol{\omega}_0 + \mathbf{j}_0 + \mathbf{h}_0$, and $\mathbf{T}_0 = \mathbf{0}$ because the state is static.
- **Prescribed free wobble:** choose $\boldsymbol{\omega}_0$, compute $\mathbf{j}_0$ from a solve with $\boldsymbol{\omega}$ held at $\boldsymbol{\omega}_0$, then $\boldsymbol{\ell}_0$ and $\mathbf{T}_0$ as above.

A load that is switched on between two time steps needs no special treatment: $\boldsymbol{\ell}$ is continuous, and the scheme produces the jump in $\boldsymbol{\omega}$ described in §8.

8 Test cases

All numbers below were checked with the reduced model eq. (10), using the `pyslfp` constants and the present-day K of N eq. 21, against a direct implementation of the step.

1. Free wobble period. From §4:

	$\dot{\mathbf{j}}$ dropped (N)	$\dot{\mathbf{j}}$ kept	difference
no ocean	444.02 d	444.48 d	+0.46 d
present-day ocean	476.03 d	476.60 d	+0.56 d

A simulation with 1-day steps reproduced the ocean value to 0.01 d.

2. Step load. If $\mathbf{j}^\zeta$ jumps by $\Delta\mathbf{j}^\zeta$, continuity of $\boldsymbol{\ell}$ gives an instantaneous $\Delta\boldsymbol{\omega} = -[I_1\mathsf{1} + (I_3 - I_1)\mathsf{K}]^{-1}\Delta\mathbf{j}^\zeta$, after which $\boldsymbol{\omega}$ circles the new static answer. For $\mathsf{K} = \kappa\mathsf{1}$ the jump is $-(I_3 - I_1)(1 - \kappa)/(I_1 + B) = -0.225\%$ of the eventual static response, in the opposite direction. The scheme reproduced this to seven significant figures.

3. Periodic forcing. For $\mathsf{K} = \kappa\mathsf{1}$ and $\mathbf{j}^\zeta \propto e^{i\nu t}$ (complex notation), eq. (2) gives

$$\frac{m_{\text{dynamic}}}{m_{\text{static}}} = \frac{1 + \nu/\Omega}{1 - \nu/\sigma'_{\text{CW}}}, \tag{20}$$

compared with $1/(1 - \nu/\sigma_{\text{CW}})$ when $\dot{\mathbf{j}}$ is dropped. The extra numerator is the $\mathsf{A}\dot{\sigma}$ term.

	prograde			retrograde		
Forcing period	30 d	182.6 d	365.25 d	30 d	182.6 d	365.25 d
$\dot{\mathbf{j}}$ kept	−0.0748	−0.701	−4.623	0.0611	0.290	0.450
$\dot{\mathbf{j}}$ dropped	−0.0725	−0.699	−4.637	0.0633	0.291	0.451

(no ocean). The forced responses differ by about ν/Ω: 3% at monthly periods and 0.3% at annual. Simulations reproduced the formula to 10^{-3} at 30 d and 5×10^{-5} at one year.

4. Long runs. The bigger practical difference is the free-wobble period. In a 20-year run with weekly steps, annual forcing and the present-day ocean, the two schemes differed by up to 6% of the response. This is mostly the slow drift in phase of the free wobble caused by the 0.56 d period difference, not the forced response.

5. Angular-momentum budget. Eq. (17) holds to rounding (2×10^{-16} relative) with non-zero $\mathbf{h}$ and $\boldsymbol{\Gamma}$.

6. Static limit. For slowly varying loads, $\boldsymbol{\omega} \to (I_3 - I_1)^{-1}(\mathbf{j} + \mathbf{h})$, i.e. the static `pyslfp` solution with `angular_momentum_change` $= -\mathbf{h}$.

9 Next steps

- **Fluid core:** add $\mathbf{h}_{\text{core}} \approx (A_f/\Omega)\mathsf{E}\dot{\boldsymbol{\omega}}$ to $\mathbf{h}$ (N§4.1). In the $\boldsymbol{\ell}$ form this replaces I_1 by A_m in $\boldsymbol{\ell}$ and in P.
- **Damping:** add a dissipative term, as in N§8.1, written for the $\boldsymbol{\ell}$ form.
- **Generalised and adjoint problems:** the $\boldsymbol{\ell}$ form keeps the structure of N§10. The adjoint is the same solver run backwards in time with the rotation reversed.